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Title: Status Epilepticus in Children

Introduction:

Status epilepticus is a medical emergency and most common neurological emergency in childhood and the incidence of status epilepticus is 18–23 per 100,000 children per year. Management of SE is aimed to stop seizures mainly to avoid cerebral damage and other morbidity.

Definition:

Traditional: Single seizure or two or more sequential seizures without return to normal consciousness/baseline in between lasting for > 30 minutes.

Operational: GTC seizure > 5 minutes or prolonged febrile seizure of 10 – 15 minutes.

This operational definition has arrived as most seizures > 5 minutes end up getting medical treatment and in children < 5 years especially if febrile, a longer timeframe of 10 to 15 minutes prior to intervention is considered appropriate.

Classification:

Early SE : 5 – 30 minutes.

Established SE : > 30 minutes.

Refractory SE : Continued seizure after 2 or 3 anticonvulsants at adequate doses.

Aim of management is to:

- 1) Administration of anticonvulsants to terminate the seizure.
- 2) Treatment of underlying precipitant causes.
- 3) Identification and management of systemic complications that could result in secondary brain injury.

Immediate management:

After giving each drug look for airway and breathing. Always provide oxygen and keep suction nearby. Watch for neurogenic pulmonary edema or respiratory depression; which is common if > 2 doses of benzodiazepines given.

- Intubation if needed.
- Monitoring O₂ saturation, blood pressure heart rate, temperature.
- Check blood glucose. (Never forget this!)
- Peripheral IV access.
- Treat for fever aggressively with goal rectal T 36 – 37 c.
- Medical and neurologic examination as possible.

Labs: U&E, magnesium, calcium, phosphate, CBC, LFT, coagulation tests, blood gas, anticonvulsant levels, if febrile blood culture

Emergent Initial Therapy (given immediately):

- If IV :
 - Diazepam 0.2mg/kg (max 10 mg) repeat after 5 minutes if no response with first dose.
 - Midazolam 0.1mg/kg (max 5 mg) repeat after 5 minutes if no response with first dose.
- If no IV:
 - Diazepam Rectal 2–5 years 0.5 mg/kg, 6–11 years 0.3 mg/kg, ≥12 years 0.2 mg/kg (max 20 mg).
 - Midazolam: IM 0.2 mg/kg (max 5 mg), buccal 0.3 mg/kg (max 5 mg).

Consider whether out-of-hospital benzodiazepines have been administered when considering how many doses to administer.

Urgent management:

CT head for status epilepticus as first presentation mandatory (febrile and non-febrile), after controlling seizure or intubation.

Additional diagnostic testing as indicated: LP, CT, MRI, toxicology labs, inborn errors of metabolism.

Consider EEG monitoring (evaluate for psychogenic status epilepticus or persisting EEG-only seizures).

Neurologic consultation is required.

Urgent control therapy:

- Phenytoin 20mg/kg IV (max 750 mg) infusion over 20 minutes under cardiac monitoring as it may cause arrhythmia, hypotension, purple glove syndrome.
- If phenytoin fails give Phenobarbital 20 mg/kg IV may cause respiratory depression and hypotension.

Refractory status epilepticus (Inform PICU consultant):

- If seizure continues after benzodiazepines and a second anti-seizure medication, the patient is in refractory status epilepticus regardless of elapsed time.
- Continue management and make plans for PICU admission/transfer.
- Once admitted in PICU consider continuous EEG monitoring once clinically evident seizure terminates to evaluate for persisting EEG-only seizures.
- Administer another urgent control anticonvulsant or proceed to pharmacologic coma.

- o Levetiracetam 20-30 mg/kg IV at 5 mg/kg/min (maximum, 3 g).
- o Valproate Sodium The initial bolus 20 mg/kg at 5 mg/kg/min. If seizures are terminated, then a continuation of periodic (twice per day) dosing may be appropriate. If seizures continue, a continuous infusion of 5 mg/kg/hr

- **Pharmacologic coma medications (To be discussed with pediatric neurologist on call):**

- o Pentobarbital: The initial bolus is 5 mg/kg followed by an infusion of 1 mg/kg/hr, which can increase as needed to 3 mg/kg/hr. Continuous blood-pressure monitoring is important

- **Pharmacologic coma management:**

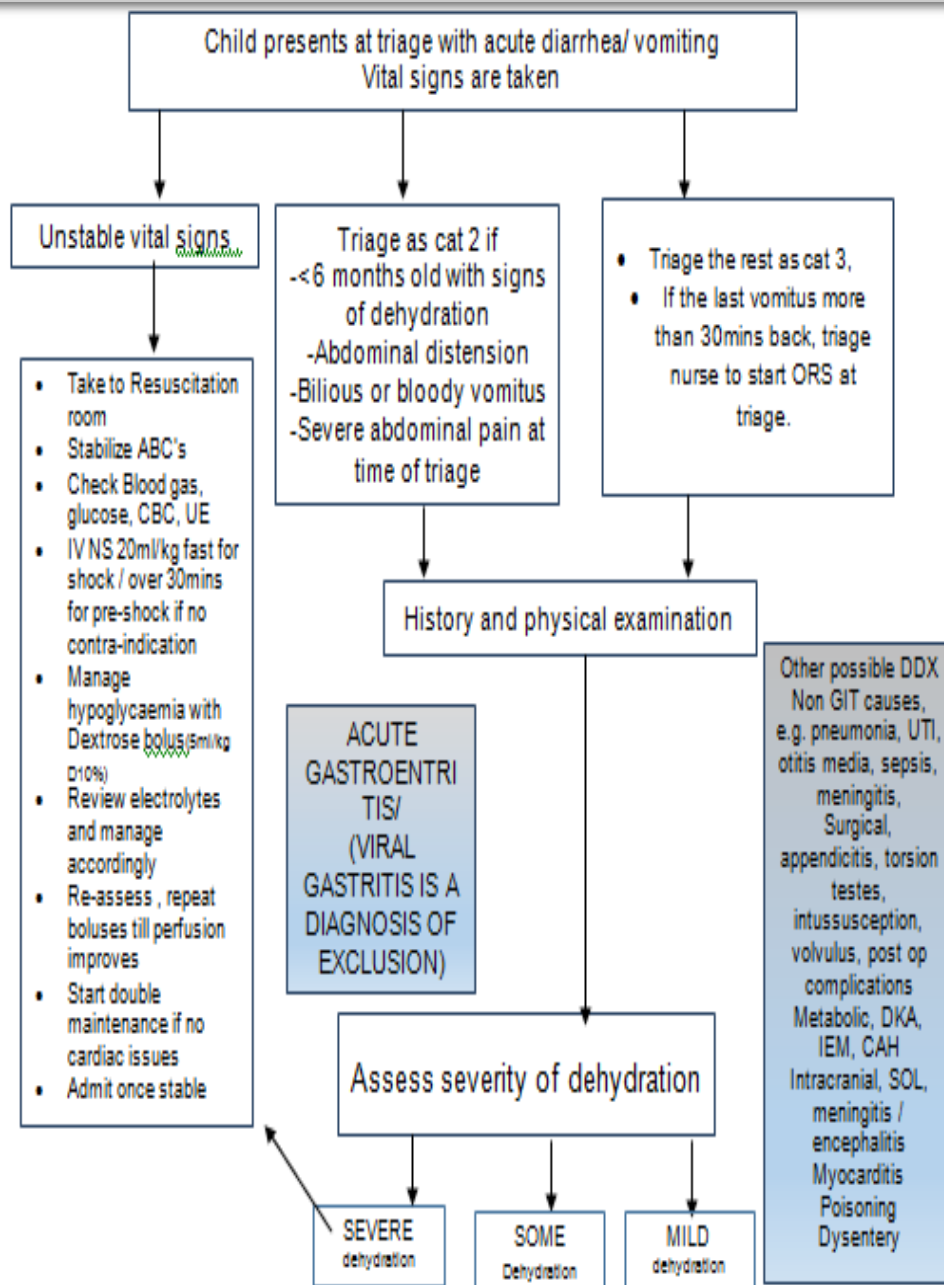
- o Titrate together seizure suppression or burst suppression based on EEG monitoring.
- o Continue pharmacologic coma for 24–48 hours.
- o Modify anti-seizure medications so additional seizure coverage is in place for infusion wean.
- o Continue diagnostic testing and implementation of etiology directed therapy.

- **Add-on options:**

Medications: phenytoin, phenobarbital, levetiracetam, valproate sodium, topiramate, lacosamide, ketamine, pyridoxine [Trial of an intravenous pyridoxine (vitamin B6, 100 mg, from 1-3 doses)] (if less than 6 months old not responding to conventional anti-epileptic, consider Pyridoxine IV, ketogenic diet, corticosteroid (**Consider neurology consultation once seizure is not under control**))

Time	Intervention
0 minute	If IV : Diazepam 0.2 mg/kg (max 10 mg) Midazolam 0.1 mg/kg (max 5 mg) If no IV: Diazepam Rectal 2–5 years 0.5 mg/kg, 6–11 years 0.3 mg/kg, ≥12 years 0.2 mg/kg (max 20 mg). Midazolam: IM 0.2 mg/kg (max 5 mg), buccal 0.3 mg/kg (max 5 mg). Consider whether out-of-hospital benzodiazepines have been administered when considering how many doses to administer.
	If IV : - If no response with first dose of Diazepam, repeat after 5 minutes 0.2 mg/kg (max 10 mg). - If no response with first dose of Midazolam, repeat after 5 minutes 0.1 mg/kg (max 5 mg). If no IV:

5 minute	- If no response with first dose of Midazolam, repeat after 5 minutes: IM 0.2 mg/kg (max 5 mg), buccal 0.3 mg/kg (max 5 mg).
10 minute	- Phenytoin 20 mg/kg IV (max 750 mg) infusion over 30 minutes to 1 hour under cardiac monitoring as it may cause arrhythmia, hypotension, purple glove syndrome. - If phenytoin fails give Phenobarbital 20 mg/kg IV may cause respiratory depression and hypotension.
	- If seizure continues after benzodiazepines and a second anti-seizure medication, the patient is in refractory status epilepticus regardless of elapsed time. - Continue management and make plans for PICU admission/transfer.



Strategy for high-dose intravenous midazolam in refractory SE		
Timing from Start of this Strategy	Midazolam Dosing	Steps
0 min: initial bolus	Give 0.5 mg/kg	A
0 min: start continuous infusion	Start at 2 $\mu\text{g/kg/min}$ (0.12 mg/kg/h)	B
5 min: if seizure persists 5 min after bolus (step A)	Give 0.5 mg/kg Increase infusion to 4 $\mu\text{g/kg/min}$ (0.24 mg/kg/h)	C
10 min: if seizure persists or recurs 5 min after step C	Give 0.1 mg/kg Increase infusion by 4 $\mu\text{g/kg/min}$ (0.24 mg/kg/h)	D
15 min: if seizure persists or recurs 5 min after step D	Repeat step D	E
20–45 min: if seizure persists or recurs 5 min after step E, then continue to repeat step D every 5 min until a maximum infusion rate is achieved	Maximum infusion rate of 36 $\mu\text{g/kg/min}$ (1.92 mg/kg/h)	F (5 cycles of D-to-E may be needed)
45 min: by the completion of step F an EEG should be available to confirm seizure control or otherwise. If seizure is not controlled then consider this episode as treatment failure and move to step H	Maintain dose of continuous infusion that achieves clinical and EEG seizure control and 24–48 h later move to step I	G
45–60 min: treatment failure	Discontinue midazolam infusion and start general anesthesia with pentobarbital or isoflurane	H
24–48 h: if patient is free of clinical and EEG seizures then start to wean the infusion	Reduce infusion by 4 $\mu\text{g/kg/min}$ every 6–8 h	I
48–84 h: continued EEG monitoring to observe for breakthrough seizures	Plan to discontinue infusion after optimizing other AEDs. If seizures recur then consider step K for weaning failure; alternatively this episode may be in the category of super-refractory SE	J
Weaning failure	Consider rebolus of 0.1 mg/kg and increase infusion by 4 $\mu\text{g/kg/min}$ and/or Consider alternative AEDs	K

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Pediatric status epilepticus management, Current Opinion Pediatric	Nicholas, S. Abend and Tobias Loddenkemper	December 2014	668–674
Pediatric Intensive Care Treatment of Uncontrolled Status Epilepticus	Ryan Wilkes, Robert C. Tasker	2013	239-257

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